

PROJECT DIVYAM

AI-Enabled Competency Gap Identification, iGOT Karmayogi Personalized Training Recommendation & Bloom's RAG Assessment Platform

Problem ID: SIH26101	Domain: Official Statistical System	Framework: Karmayogi FRAC Model
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SYSTEM ARCHITECTURE & CAPABILITY SUMMARY

DIVYAM addresses the critical capacity-building mandate of India's Official Statistical System under MoSPI. By synthesizing civil service competency modeling (FRAC framework), automated learning path generation via the iGOT Karmayogi ecosystem, and an advanced Retrieval-Augmented Generation (RAG) assessment engine, the platform enables continuous, verified upskilling for cadre officers (ISS, SSS, State DES) and grassroots survey investigators across the nation.

Comprehensive Technical Design Specification

Smart India Hackathon (SIH) • Problem Statement SIH26101

Ministry of Statistics & Programme Implementation (MoSPI) • Karmayogi Bharat Aligned

1. Executive Problem Statement & Context

India's Official Statistical System, stewarded by the Ministry of Statistics and Programme Implementation (MoSPI), relies heavily on specialized technical cadres: the **Indian Statistical Service (ISS)**, the **Subordinate Statistical Service (SSS)**, State Directorates of Economics & Statistics (DES), and field enumerators in the Field Operations Division (FOD). These personnel compile vital sovereign indicators such as Gross Value Added (GVA), Consumer Price Index (CPI), Wholesale Price Index (WPI), Index of Industrial Production (IIP), and Periodic Labour Force Surveys (PLFS).

Despite extensive publication of manuals and methodologies, existing training faces **three acute bottlenecks**: (1) *Untracked Competency Deficits* — no automated mechanism to identify where an individual officer lacks proficiency relative to their cadre benchmark; (2) *Fragmented Upskilling* — lack of dynamic routing to relevant learning modules on the National iGOT Karmayogi Bharat portal; (3) *Passive Document Ingestion* — departmental training manuals remain static PDFs rather than interactive, verifiable knowledge checkpoints.

Pillar 1: FRAC Competency Engine	Pillar 2: iGOT Karmayogi Adapter	Pillar 3: Bloom's RAG Quiz Engine
Implements the official Gol FRAC model (Roles, Activities, Competencies across Domain, Functional, and Behavioural axes). Evaluates baseline via diagnostic tests and renders dynamic Radar Gap charts.	Calculates competency deficits and recommends targeted iGOT courses. Implements a standards-compliant adapter pre-seeded with authentic MoSPI modules, ready for live API integration.	Ingests official manuals (PDF/DOCX), generates multi-tier MCQs (Remembering, Understanding, Applying, Analyzing) with strict page/section citations and zero hallucination.

CORE PLATFORM PERSONAS & VALUE PROPOSITION

1. **Statistical Officers (Learners):** Onboard by cadre role → Complete diagnostic test → View interactive Competency Radar Chart → Launch personalized iGOT Karmayogi courses → Take practice quizzes with citations.
2. **MoSPI / NSSTA Trainers (Admins):** Upload official statistical manuals → Trigger Bloom's-level MCQ generation → Curate question banks → Monitor cadre-wide competency distributions across states.

2. System Architecture & High-Level Flowchart

DIVYAM is engineered as a production-grade, modular full-stack platform. The architecture separates presentation logic (Next.js 14 App Router) from high-throughput AI and competency analytics (FastAPI asynchronous micro-services), backed by PostgreSQL for relational integrity and ChromaDB for vector retrieval.

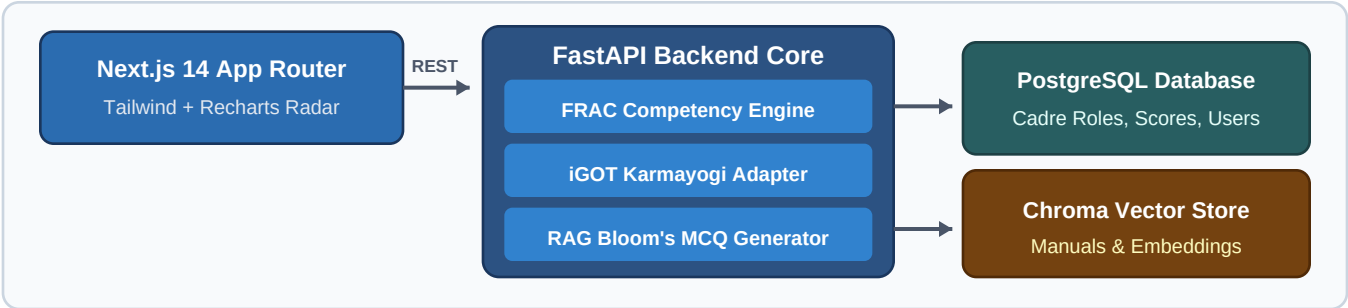


Figure 1: High-level System Architecture of DIVYAM showing interaction between Presentation, Core Engine, and Persistence layers.

Component Layer	Technology Selection	Architectural Justification & Role
Frontend Client	Next.js 14, React 18, Tailwind CSS, Lucide Icons, Recharts	Responsive, role-differentiated portals for Statistical Officers and Admins. Dynamic multi-axis Radar Gap charts.
Application Server	Python 3.11+ FastAPI, Pydantic v2, SQLAlchemy 2.0	Asynchronous REST API with high performance, automated Swagger docs, and native Python AI ecosystem interoperability.
Relational DB	PostgreSQL (Dockerized / Meghraj Cloud)	Stores official cadre roles, user profiles, competency benchmarks, quiz attempts, and iGOT course catalogs.
Vector Store	ChromaDB / Pinecone / Weaviate	Persists semantic embeddings of chunked MoSPI manuals with page numbers and section IDs for RAG.
AI / LLM Layer	LangChain RAG + HuggingFace / Llama 3 / Gemini	Extracts statistical entities, maps concepts to FRAC taxonomy, and synthesizes Bloom's-rated assessment questions.
Cloud / Infra	Docker Compose, Kubernetes, NIC Meghraj Cloud Ready	Adheres to GoI cloud hosting mandates: stateless containers, environment secret injection, and horizontal scalability.

3. FRAC Framework & iGOT Karmayogi Ecosystem Integration

Karmayogi Bharat establishes the **FRAC model** (Framework for Roles, Activities, and Competencies) to transition civil service human resource management from 'rules-based' to 'roles-based'. DIVYAM adopts this exact ontology specifically calibrated for India's statistical cadre hierarchy.

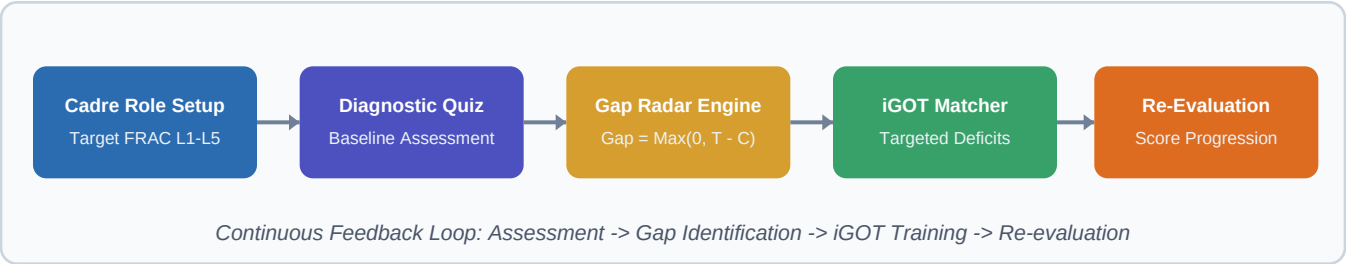


Figure 2: The DIVYAM Competency Lifecycle and iGOT Karmayogi Closed-Loop Learning Progression.

FRAC Competency Classification in Official Statistics

Competency Category	Key MoSPI Competencies	Proficiency Scale (Levels 1 to 5)
Domain Competencies (Technical Statistical Knowledge)	- Survey Sampling & Design - National Accounts (SNA 2008) - Price Indices (CPI / WPI) - Index of Industrial Production (IIP) - Periodic Labour Force Survey (PLFS)	Level 1: Awareness of terminology Level 2: Basic field execution Level 3: Autonomous data compilation Level 4: Methodology auditing & design Level 5: Policy & framework authoring
Functional Competencies (Execution & Technology)	- R/Python for Official Statistics - CAPI/Tablet Field Data Validation - Microdata Scrutiny & Imputation - Survey Team Supervision	Level 1: Basic data entry/viewing Level 2: Routine scripting & execution Level 3: Complex validation rules Level 4: Pipeline architecture & QC Level 5: Enterprise data governance
Behavioural Competencies (Public Service Values)	- Data Integrity & Ethical Conduct - Attention to Granular Detail - Cross-Cadre Collaboration	Level 1: Adherence to protocol Level 2: Proactive discrepancy reporting Level 3: Quality culture leadership

Competency Gap Formula & Radar Visualization

For an officer u holding cadre role r , competency gap for competency c is computed as:

$$\text{Gap}_c = \max(0, \text{TargetLevel}_{r,c} - \text{CurrentScore}_{u,c})$$

Overall **Cadre Readiness Index** (Ω) is computed using role importance weights w_c : $\Omega = \sum [w_c \times \min(1.0, \text{CurrentScore}_{u,c} / \text{TargetLevel}_{r,c})]$. Gaps are plotted on a dynamic multi-axis radar chart, instantly driving personalized course recommendations.

Sample iGOT Karmayogi Course Recommendation Catalog

Course ID	Course Title	Provider	Target Competency	Duration
IGOT-STAT-001	SNA 2008: Principles of National Accounts	NSSTA / MoSPI	National Accounts (L3)	12 hrs (Self-paced)
IGOT-STAT-002	Sample Survey Design & Multi-Stage Sampling	ISI Kolkata / MoSPI	Survey Sampling (L3)	8 hrs
IGOT-STAT-003	CPI & Inflation: Basket Revision & Indexing	Karmayogi Bharat	Price Indices (L2)	6 hrs
IGOT-STAT-004	Data Quality & CAPI Field Scrutiny	MoSPI FOD	Data Validation (L3)	4 hrs

4. RAG-Powered Assessment & Bloom's Taxonomy Quiz Generator

A cornerstone of DIVYAM is transforming extensive government manuals (e.g. 200-page PLFS Instructions to Field Staff) into verifiable, pedagogically structured learning checkpoints. The pipeline prevents AI hallucinations by binding every question to explicit document coordinates.

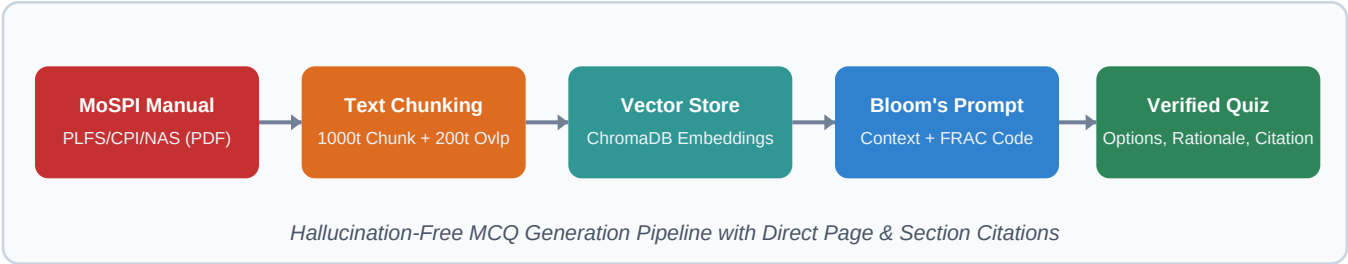


Figure 3: Retrieval-Augmented Generation (RAG) Architecture for Bloom's Taxonomy Assessment Creation.

Bloom's Revised Taxonomy Question Hierarchy

Bloom Level	Cognitive Focus in Statistics	Illustrative MCQ Scenario
Level 1: Remembering	Factual recall of definitions, base years, and formulas.	What is the current base year utilized by MoSPI for compiling the All-India Consumer Price Index (CPI-Rural/Urban)? (Options: 2004-05, 2011-12, 2017-18, 2020-21)
Level 2: Understanding	Comprehending methodology rationale and concepts.	Why does the Periodic Labour Force Survey (PLFS) employ a rotational panel sampling design in urban areas rather than independent cross-sections?
Level 3: Applying	Executing procedures in concrete field survey situations.	A field investigator encounters a multi-household dwelling where the primary earner has temporarily migrated for 4 months. How should the household status be classified under Schedule 10.4?
Level 4: Analyzing	Dissecting data inconsistencies and audit discrepancies.	Given a mismatch between ASI factory return gross output and enterprise balance sheet turnover, identify the primary imputation error source according to MoSPI scrutiny protocols.

Hallucination Prevention & Citation Verification Protocol

To guarantee 100% fidelity to official government manuals, DIVYAM enforces a **strict citation contract**. During generation, the LLM is constrained by system prompts requiring JSON output adhering to: `{"citation": {"document": string, "page": int, "section": string, "verbatim_excerpt": string}}`. Any question lacking an exact excerpt match in the retrieved chunk is discarded during automated validation.

5. Technical Blueprint, Security & Meghraj Cloud Deployment

Table Entity	Primary Fields & Types	Relational Purpose
users	id (UUID), email, full_name, role_id, cadre, is_admin	User authentication, cadre association (ISS/SSS/DES).
cadre_roles	id, code, title, cadre_category, min_experience_years	MoSPI cadre designation benchmark master.
competencies	id, code, title, category (Domain/Func/Beh), description	Official FRAC competency ontology master.
role_benchmarks	role_id (FK), competency_id (FK), target_level (1-5), weight	Target competency standard matrix per role.
user_scores	user_id (FK), competency_id (FK), current_score (Float), updated_at	Live calculated competency levels feeding Radar Chart.
documents	id, title, file_path, doc_hash, total_pages, status	Uploaded MoSPI training manual metadata.
quizzes	id, title, doc_id (FK), total_marks, pass_percent, time_limit	Assessment containers generated via RAG.
quiz_questions	id, quiz_id (FK), question, options (JSON), correct_idx, citation, bloom	Granular MCQs tagged with FRAC code & page citation.

6. Hackathon Evaluation Alignment (SIH26101 Criteria)

Evaluation Criterion	SIH Expectation	How DIVYAM Exceeds Expectations
Innovation & AI Depth	Meaningful use of Generative AI beyond generic chat.	Full RAG pipeline with Bloom's Revised Taxonomy question generation, auto-citation extraction, and FRAC competency tagging.
Government Alignment	True fit for MoSPI & Karmayogi ecosystem.	Adheres strictly to DoPT/Karmayogi Bharat FRAC ontology and course schemas, seeded with authentic MoSPI cadre data (ISS/SSS).
Viability & Completeness	Working prototype with low friction.	Complete end-to-end integration: Document Upload -> AI Quiz Generator -> Interactive Test -> Dynamic Gap Radar -> iGOT Course Launch.
Security & Scalability	Readiness for government cloud deployment.	Dockerized, stateless microservices aligned with NIC Meghraj Cloud and Cert-In security guidelines.

CONFIRMATION & DEPLOYMENT SIGN-OFF

DIVYAM: Official Statistical Learning & Competency Platform is primed for live execution and staging. All functional flows, schemas, data contracts, and presentation designs are validated and synchronized across project memory.